

Lecture 23.01: Introduction to Machine Learning

 Python 3.10+

 Scikit-Learn 1.4+

Status Completed

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1. What is Machine Learning?

Machine Learning (ML) is a subfield of Artificial Intelligence that gives computer systems the ability to automatically learn patterns from data and improve their performance on a specific task without being explicitly programmed with deterministic hand-crafted rules.

Historical & Formal Definitions:

Arthur Samuel (1959):

"Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed."

Tom Mitchell (1997) — Formal Engineering Definition:

"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ."

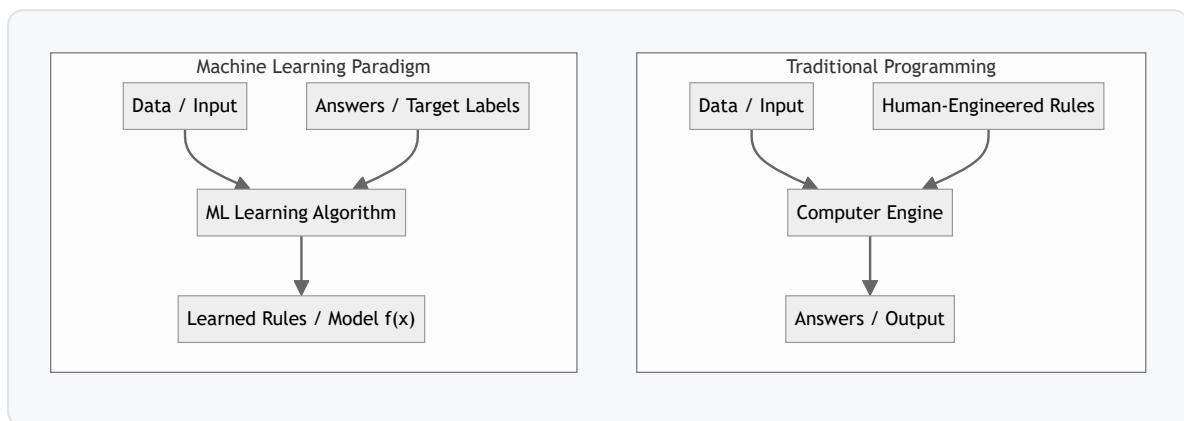
Breakdown of the (T, P, E) Framework:

1. **Task (T):** The operational problem to be solved (e.g., predicting insurance charges, filtering spam emails, recognizing handwritten digits).
2. **Performance Measure (P):** A quantitative metric evaluating how well the system performs the task (e.g., Mean Squared Error, Accuracy, R^2 Score).

3. **Experience (E):** Historical observation data supplied to the algorithm (e.g., tabular records, customer transactions, images).

2. Traditional Programming vs. Machine Learning Paradigm

In classical software engineering, humans write explicit logical conditions (if-else branching, loops). In Machine Learning, the system infers the mapping function $f : \mathbf{x} \mapsto \mathbf{y}$ directly from inputs and outputs.



3. Why Machine Learning is Necessary

Traditional rule-based systems collapse when faced with:

1. **High Dimensionality:** In images with $1024 \times 1024 \times 3 \approx 3.14 \times 10^6$ pixels, manual if-else rules are humanly impossible to construct.
 2. **Dynamic Non-Stationarity:** Spam patterns, fraud techniques, and user behavior mutate constantly; manual rules require endless updates.
 3. **Hidden Latent Patterns:** In genomic sequencing and financial time series, patterns are governed by complex non-linear probabilistic interactions that cannot be deduced analytically.
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4. Python Demonstration: Rule-Based vs. Learned Model

```
import numpy as np

# Synthetic feature: Square footage of house (in 1000 sq ft)
X = np.array([1.0, 1.5, 2.0, 2.5, 3.0])
# Target: House price (in $100k)
y = np.array([2.1, 2.9, 4.1, 5.0, 5.9])

# Traditional approach: Hard-coded human guess
def rule_based_predict(x):
    return 2.0 * x # Static hand-crafted rule

# Machine Learning approach: Estimate optimal parameters w, b
# Using closed-form Least Squares formula:  $w = \text{Cov}(X,y)/\text{Var}(X)$ 
X_mean, y_mean = np.mean(X), np.mean(y)
w_learned = np.sum((X - X_mean) * (y - y_mean)) / np.sum((X - X_mean)**2)
b_learned = y_mean - w_learned * X_mean

print(f"Human Hard-coded Rule:  $y = 2.0 * x$ ")
print(f"Learned ML Model:  $y = \{w\_learned:.3f\} * x + \{b\_learned:.3f\}$ ")
```